

which we assume have an interest rate of zero, and equities. The SWF has CRRA utility (which is the academic version of mean-variance utility; see chapters 2 and 4) and decides on the amount of equities to hold in its portfolio in each period. These are given by α_1 and α_2 , respectively:

$$\max_{\alpha_1, \alpha_2} E \left[\frac{W_2^{1-\gamma}}{1-\gamma} \right], \quad (5.4)$$

where γ is the nation's degree of risk aversion, which I set to 10. Equation (5.4) states that we want to maximize the country's expected utility, where the expression in the brackets is CRRA utility over terminal wealth, and we do that by finding the optimal period 1 and 2 portfolio weights. I constrain the portfolio weights to lie between 0 and 1 (so there can be no shorting).²³

Oil is like labor income. The SWF can harvest oil of L_t at the beginning of each period, and fluctuating oil prices generating a return at the end of the period. Oil income reduces by 90% from period 1 to period 2, so $L_1 = 0.1 \times L_0$. Calibrating this model using data from 1983 to 2012, the correlation of oil returns with equities (S&P 500) is a low 4%. (This might seem surprising, and it turns out that commodities are also not very good inflation hedges, as chapter 11 discusses.)

Figure 5.5 graphs the asset allocations to equity as a function of L/W , the ratio of oil to financial wealth at the beginning of each period. Panel A shows the allocations to equities at the empirical correlation of 4%. In each period, the allocations to equity slope upward. As there is more oil, the SWF holds more equities because of diversification. Equities are lowly correlated with oil, so there is large scope for equities to pay off when oil does not, making equities increasingly attractive when oil reserves are large. Notice that the first period equity holdings are larger than the second period for a given L/W ; the SWF reduces equities with time because oil is "bond-like," which in this case means oil is minimally correlated with equities. The SWF is endowed with a lot of bond-like securities at the first period, so holds more equities. In period 2, oil is mostly extracted, and so the SWF lowers the equity weight to counter the disappearing bond-like oil income.

In Panel B of Figure 5.5, I contrast the case where oil returns and S&P 500 returns are positively correlated at 0.3. In this case, the equity shares are decreasing in L/W and the period 0 and 1 weights flip. The SWF holds fewer equities for a given level of L/W in the second period because oil is now stock-like. In the first period, the SWF is endowed with a stock-like oil resource. After most of the oil is extracted, the SWF wishes to replace that stock-like exposure to maintain the same overall risk profile, so it increases equity holdings in the second period.

²³ For those who are technically adept: the case of power utility, i.i.d. log-normal returns for asset returns and labor income is not analytical. See Duffie et al. (1997) and Koo (1998). All interesting life-cycle models are solved numerically.